Lumina AI – Final Demonstration Plan

Multi-domain demonstration of ingestion, grounded answers, clarification, multi-turn conversation, voice, transparency and analytics

> Items marked **[Record]** are blanks to fill from the rehearsal and the final run. Only report what you actually observed.

1. Demonstration Goals

Show, using three loaded knowledge domains, that Lumina AI can:

- ingest documents in several formats;
- resolve queries with source-grounded answers;
- ask for clarification when a question is ambiguous;
- hold a multi-turn conversation with correct context and topic switching;
- accept voice input and speak answers;
- expose transparency information (sources, snippets, scores, confidence);
- identify low-confidence and unanswered queries;
- use analytics to reveal potential knowledge gaps.

Target length: 12 to 15 minutes.

2. Preparation Checklist

Item	Action	Done
Environment	Backend on `127.0.0.1:8000`, frontend on `localhost:5173`, `/api/health` returns OK	☐
API key	Gemini key set in `.env`; one test query succeeds	☐
Browser	Chromium-based browser, microphone permission granted, speakers on	☐
Documents	2 to 3 short files per domain (Section 3), varied formats (PDF, DOCX, TXT, CSV)	☐
Clean state	Start with an empty knowledge base and empty query log so analytics are genuine	☐

Item	Action	Done
Fallback	Record a backup screen capture of a successful run in case of network failure	☐
Screenshots	Capture upload, answer with sources, clarification, voice, analytics for the report	☐

3. Knowledge Domains and Documents

Domain	Suggested content	Format
AI / NLP	Short paper or notes on RAG, embeddings, transformers	PDF
Remote Sensing / Earth Observation	Notes on SAR backscatter, NDVI, soil moisture / vegetation optical depth	DOCX or PDF
Cybersecurity	Notes on encryption, OWASP top risks, incident response	TXT
Structured data	A small CSV, for example a glossary or a table of vegetation indices	CSV

Choose real documents you are allowed to use. Do not preload answers for out-of-scope questions, so the unanswered-query demonstration is genuine.

4. Demonstration Flow

```
Intro > Ingest > Grounded answers > Clarification > Multi-turn > Voice
      > Transparency > Unanswered / low-confidence > Analytics + gaps > Close
the gap > Wrap-up
```

5. Script

Scene 1: Introduction (1 min)

Present the problem: manual search is slow, keyword search misses meaning, chatbots may not be grounded. Show the architecture diagram from the Technical Documentation and name the five agents.

Scene 2: Knowledge ingestion (2 min)

1. Open the Knowledge Base page (empty).
2. Upload one file from each domain, including the CSV.
3. Point out processing status, chunk counts and the `ready` state.
4. Optionally upload an unsupported file to show validation.
5. Say: "Each chunk now carries document, page and chunk metadata and a vector in the index."

Scene 3: Source-grounded answers across domains (3 min)

Domain	Question	What to point out
AI / NLP	"What is Retrieval-Augmented Generation?"	Factual classification; answer matches retrieved text
Remote Sensing	"What does NDVI measure and how is it calculated?"	Sources panel names the correct document and page
Cybersecurity	"How do symmetric and asymmetric encryption differ?"	Comparative classification; structured side-by-side answer
Procedural	A "how do I" question answered by a loaded document	Step-by-step answer

For each answer, open the Sources panel and show document name, snippet, relevance score and confidence badge. Remind the audience that the score measures retrieval similarity, not correctness.

Scene 4: Clarification (1 to 2 min)

1. Ask an ambiguous question: "What about memory?"
2. Show the Clarification Agent asking one focused question.
3. Reply, for example: "Conversation memory in the RAG system."
4. Show the refined query being answered with sources.

Scene 5: Multi-turn conversation (2 min)

1. "What is SAR backscatter?"
2. Follow-up without the topic: "How does it respond to vegetation?"
3. Follow-up: "And what about soil moisture?"
4. **Topic switch:** "Now, what are the steps of incident response?" Show that the answer comes from the cybersecurity document with no leakage from the SAR thread.

Scene 6: Voice interaction (1 to 2 min)

1. Click the microphone and speak an in-scope question.
2. Show the live transcript, edit if needed, then submit.
3. Enable text-to-speech and let the answer play; demonstrate pause and stop.
4. Optionally show the message displayed when microphone permission is denied.

Scene 7: Transparency (1 min)

Open a source for one answer: document, page, chunk ID, snippet, score. Optionally compare it side by side with the original document to prove the snippet is real.

Scene 8: Low-confidence and unanswered queries (2 min)

Ask questions that the loaded documents do not cover. Examples, if absent from your files:

- "What is the penalty for late tax filing?"
- "How does quantum key distribution work?" (if not in the cybersecurity notes)
- A product-specific detail that your loaded notes do not contain.
- A borderline question that only loosely touches the loaded material, to trigger a low-confidence flag.

Show that the system says it could not find the information in the available documents rather than inventing an answer, and that the low-confidence case is visibly flagged. Repeat one unanswered topic two or three times in different wording so that a gap forms.

Scene 9: Analytics and knowledge gaps (2 min)

Open the Analytics dashboard and walk through:

- Total queries, category distribution, clarification rate.
- Response-time summary.
- Most-used documents.
- **Low-confidence list** with scores.
- **Unanswered list** with the questions just asked.
- **Knowledge-gap list**: the repeated topic with counts and example questions.

Close the loop: upload a document that covers the gap, repeat the question, show it is now answered with sources, then return to analytics to show the new query was recorded as answered.

Scene 10: Wrap-up (1 min)

Summarize the benefits, state the limitations honestly (OCR, tables, model API dependence, voice browser support), and mention future enhancements.

6. Evidence Table

Complete after the rehearsal and the final run.

Capability demonstrated	Scene	Outcome	Evidence (screenshot or log)
Multi-format ingestion	2	[Record]	
Cross-domain grounded answers	3	[Record]	
Source and score display	3, 7	[Record]	
Clarification	4	[Record]	
Multi-turn memory and topic switch	5	[Record]	
Voice input and TTS	6	[Record]	
No-evidence handling	8	[Record]	
Low-confidence flagging	8	[Record]	
Analytics from real logs	9	[Record]	
Knowledge-gap detection and fix	9	[Record]	

7. Likely Questions and Suggested Answers

Question	Suggested answer
How do you prevent hallucination?	Answers are generated only from retrieved chunks, the prompt forbids unsupported claims, below-threshold retrieval returns a "not found" response, and sources are shown for checking. This reduces but does not eliminate error.
What does the confidence score mean?	It reflects retrieval quality (similarity scores and coverage), not guaranteed factual correctness.

Question	Suggested answer
Why multiple agents?	Each stage is simpler, independently testable and tunable, and failures are easier to locate.
How is memory different from the knowledge base?	Memory is temporary session context; the knowledge base is the indexed document content.
How are knowledge gaps found?	Unanswered and low-confidence queries are grouped by topic from the stored log, and recurring topics are listed.
What happens when the model API is down?	The user sees a clear error message; the failure is logged and the app does not crash.
How would it scale?	Move to a production vector database, add hybrid retrieval, caching and background ingestion workers.

8. Contingency Plan

- **Model API slow or failing:** switch to the recorded backup run and explain the dependency.
- **Microphone not working:** use typed input and show a pre-recorded voice clip.
- **Upload fails:** use a document already ingested during rehearsal.
- **Unexpected wrong answer:** open the Sources panel, explain what was retrieved, and use it to illustrate the limitation honestly.

9. Final Submission Checklist

- ☐ Technical documentation reviewed against the codebase (all [Verify] items resolved).
- ☐ Test results filled in with Passed, Failed, Warning or Not Tested (all [Record] items resolved).
- ☐ Screenshots and logs attached.
- ☐ Three-domain test results available before claiming Milestone 4 coverage.
- ☐ README, `.env.example` and run instructions in the repository.
- ☐ API key excluded from Git history.
- ☐ Demonstration rehearsed end to end at least once.